

3573. Best Time to Buy and Sell Stock V

Difficulty: Medium

Link: <https://leetcode.com/problems/best-time-to-buy-and-sell-stock-v/?envType=daily-question&envId=2025-12-17>

Generated at: 2025-12-17 23:46:51

Problem:

You are given an integer array `prices` where `prices[i]` is the price of a stock in dollars on the i^{th} day, and an integer `k`.

You are allowed to make at most `k` transactions, where each transaction can be either of the following:

- **Normal transaction:** Buy on day `i`, then sell on a later day `j` where `i < j`. You profit `prices[j] - prices[i]`.
- **Short selling transaction:** Sell on day `i`, then buy back on a later day `j` where `i < j`. You profit `prices[i] - prices[j]`.

Note that you must complete each transaction before starting another. Additionally, you can't buy or sell on the same day you are selling or buying back as part of a previous transaction.

Return the **maximum** total profit you can earn by making **at most** `k` transactions.

Example 1:

Input: `prices = [1,7,9,8,2]`, `k = 2`

Output: 14

Explanation:

We can make \$14 of profit through 2 transactions:

- A normal transaction: buy the stock on day 0 for \$1 then sell it on day 2 for \$9.
- A short selling transaction: sell the stock on day 3 for \$8 then buy back on day 4 for \$2.

Example 2:

Input: prices = [12,16,19,19,8,1,19,13,9], k = 3

Output: 36

Explanation:

We can make \$36 of profit through 3 transactions:

- A normal transaction: buy the stock on day 0 for \$12 then sell it on day 2 for \$19.
- A short selling transaction: sell the stock on day 3 for \$19 then buy back on day 4 for \$8.
- A normal transaction: buy the stock on day 5 for \$1 then sell it on day 6 for \$19.

Constraints:

- $2 \leq \text{prices.length} \leq 10^3$
- $1 \leq \text{prices}[i] \leq 10^9$
- $1 \leq k \leq \text{prices.length} / 2$